Waipapa Taumata Rau
University of Auckland

Business Case Analysis and Options Assessment for System Level Change

Lecture 4

July 28, 2026

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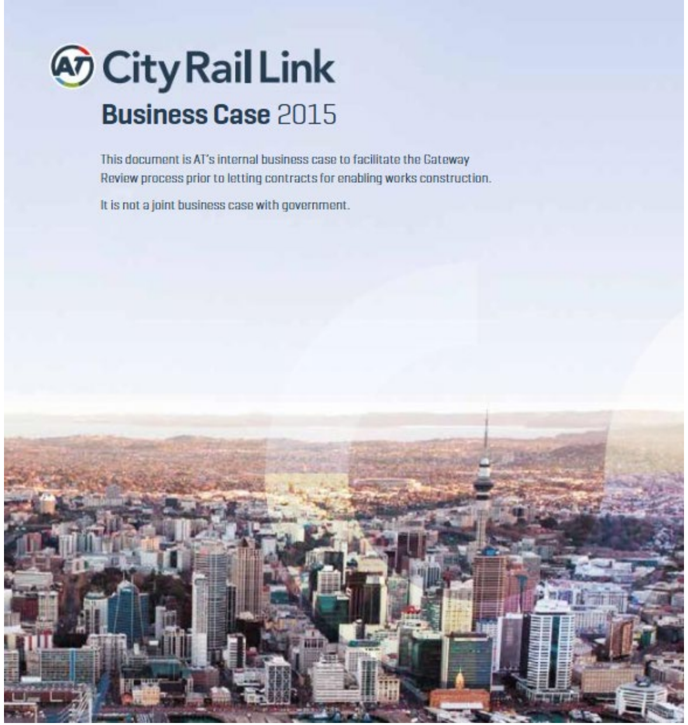
Learning outcomes

1. Explain how the Better Business Case framework supports systems-level decision making.
2. Generate and evaluate solution options using systems engineering tools.
3. Develop investment packages and justify trade-offs between alternative solutions.
4. Apply these principles to structure a systems-level business case.

Outline

1. Business case at the systems level (Better Business Case)
2. Business case methodology
3. Business case sections
4. Generating options
5. Options assessment

Business cases inform decision making



[CRL Business Case]

[1 news article]

Business cases inform decision making

New Dunedin Hospital Business Cases

Business cases

A business case:

- Justifies why a project should be undertaken.
- Defines the problem and why change is needed.
- Explores multiple options, including doing nothing.
- Evaluates benefits, costs, risks, and trade-offs of each option.
- Recommends the preferred solution based on evidence.
- Provides the justification for investment and funding.
- Builds confidence among decision-makers and stakeholders.
- Forms the basis for implementation and project delivery.

Business Cases ENGGEN 303 vs ENGGEN 403

- Product-level business cases typically focus on customers, suppliers, competitors, and commercial viability.
- Systems-level projects involve a much broader range of stakeholders, including governments, communities, iwi, businesses, regulators, and future generations, each with different priorities.
- Problems become larger in scope, spanning multiple sectors and interconnected systems where actions can have unintended consequences.
- Success is no longer measured solely by profitability or customer value, but by balancing economic, social, environmental, cultural, and long-term outcomes for society.
- Decisions must consider uncertainty, competing stakeholder perspectives, and the wider impacts on society.

Better Business Case framework

Better Business Case framework

Simple, low-risk investments may only require a single business case. Larger, more complex, or higher-risk investments typically require a staged Better Business Case process.

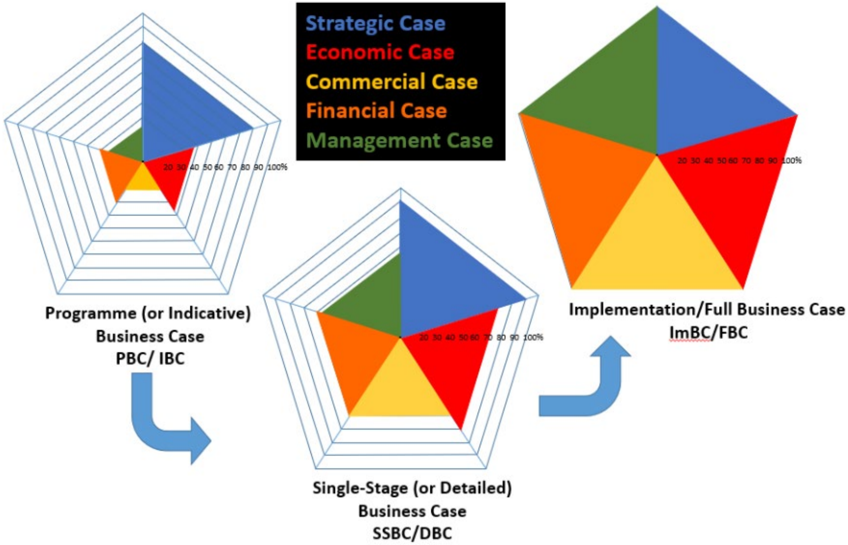
Stage 1 - Indicative Business Case (IBC)

- Why is investment needed?
- What options are available?
- Which option should be investigated further?

Stage 2 - Detailed Business Case

- Fully develop the preferred option
- Confirm costs, benefits and risks
- Seek approval to deliver the project

If approved, develop full implementation business case



Auckland LIGHT RAIL
Bringing us closer

[Auckland Light Rail Indicative Business Case](#)

[Irex Detailed Business Case](#)

Better Business Case framework – Programme Business Cases

Auckland Rapid Transit Plan:

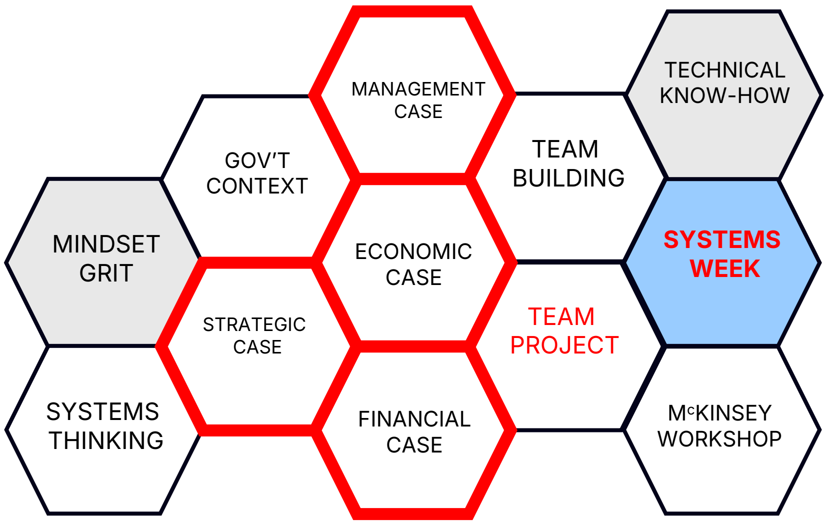
Under construction or completed

- City Rail Link (CRL) *(completed 2026)*
- Western Ring Route *(completed 2023)*
  - Waterview connection *(completed 2017)*
  - Northern corridor upgrades *(completed 2023)*
- Eastern Busway
- Northern Busway
- Airport to Botany
  - Auckland light rail (cancelled)

Planned

- Northwest Busway
- Waitemata harbour connections

Better Business Case framework to ENGGEN 403



Strategic case "The case for change"

- Project background/Introduction
- Problem space and problem statements
- Stakeholder Analysis
- Requirements and Critical Success Factors
- Key Constraints and Assumptions

Economic case "What are the options?"

- Long list options assessment
  - DFV
  - CSFs and problem statements
- Shortlisted options packages

Financial case "What are the costs and benefits?"

- Economic and social impact assessment of at least 2 shortlisted options
- Preferred way forward

Management case "How are we going to do it"

- High level Implementation and Project Plan

Business case methodology

1. Define and analyse a **problem space** and form **problem statements**
2. Identify the **users/stakeholders**
3. Determine the **requirements** of the users/stakeholders
4. Select the **Critical success factors (CSFs)** that a solution must meet
5. Generate different **options** that could address the problem
6. Assess options against the **DFV, CSFs** and problem statements to narrow down options
7. **Package** options together based on scale, and options assessment
8. Carry out **economic analysis and societal impact** of the shortlisted options
9. Based on this analysis suggest a **preferred way forward**
10. Outline a **plan to implement** the preferred way forward

Systems level business cases

SYSTEMIC INTERVENTIONS FOR PLASTIC POLLUTION IN AOTEAROA

ENGGEN 403 SYSTEMS THINKING

OCTOBER 3 2025

Project Background

Project background should answer three questions

- What is the problem or opportunity?
- Why does it matter (impact)?
- How did we get here? aka What is the status quo?

For the past 75 years, plastic has been mass produced becoming a part of everyday life due to its durability, lightweight nature, and cost-effectiveness (United Nations Environment Programme, 2021). These properties make it ideal for medical equipment, technological devices, and even disposable bags (Strategy for Plastics Innovation, 2023). However its extensive use has imposed significant environmental costs, with trillions of plastic fragments now present in marine ecosystems (Center for Biological Diversity, 2019). Studies have also shown significant impacts on human health with microplastics often found in human bodies, highlighting significant issues with New Zealand's current plastics use and handling (Pantos & Gerrard, 2024).

Business cases

ENGGEN 303

B2C

B2B

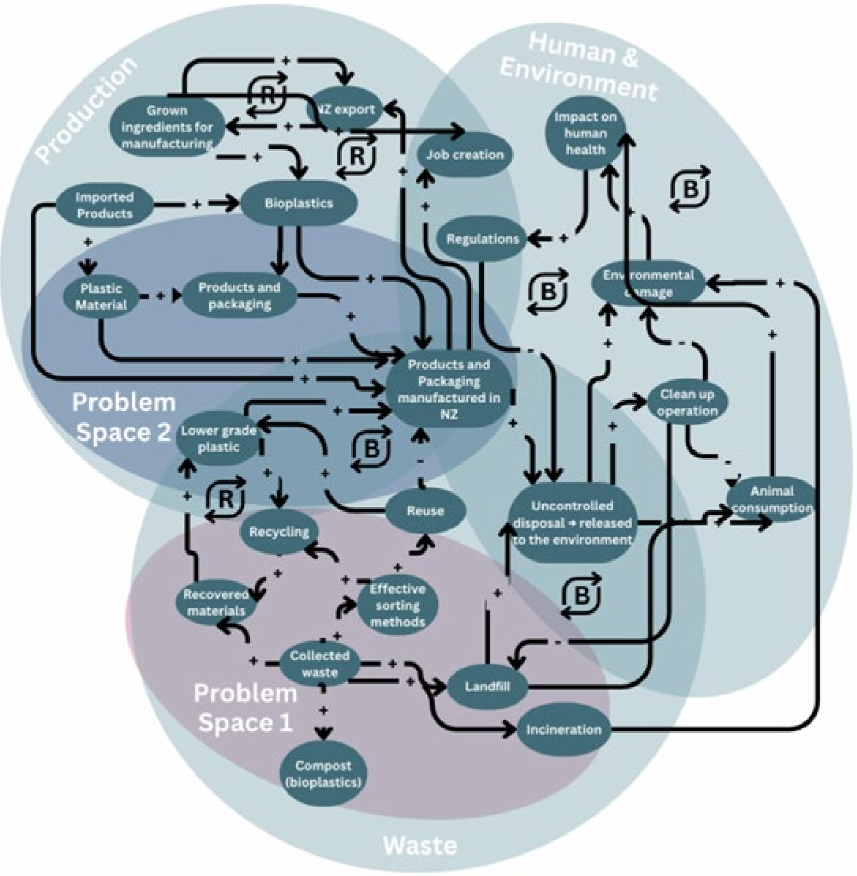
Simple to complicated problems

ENGGEN 403

B2G

Complex and Wicked problems

Lecture 8 – next week



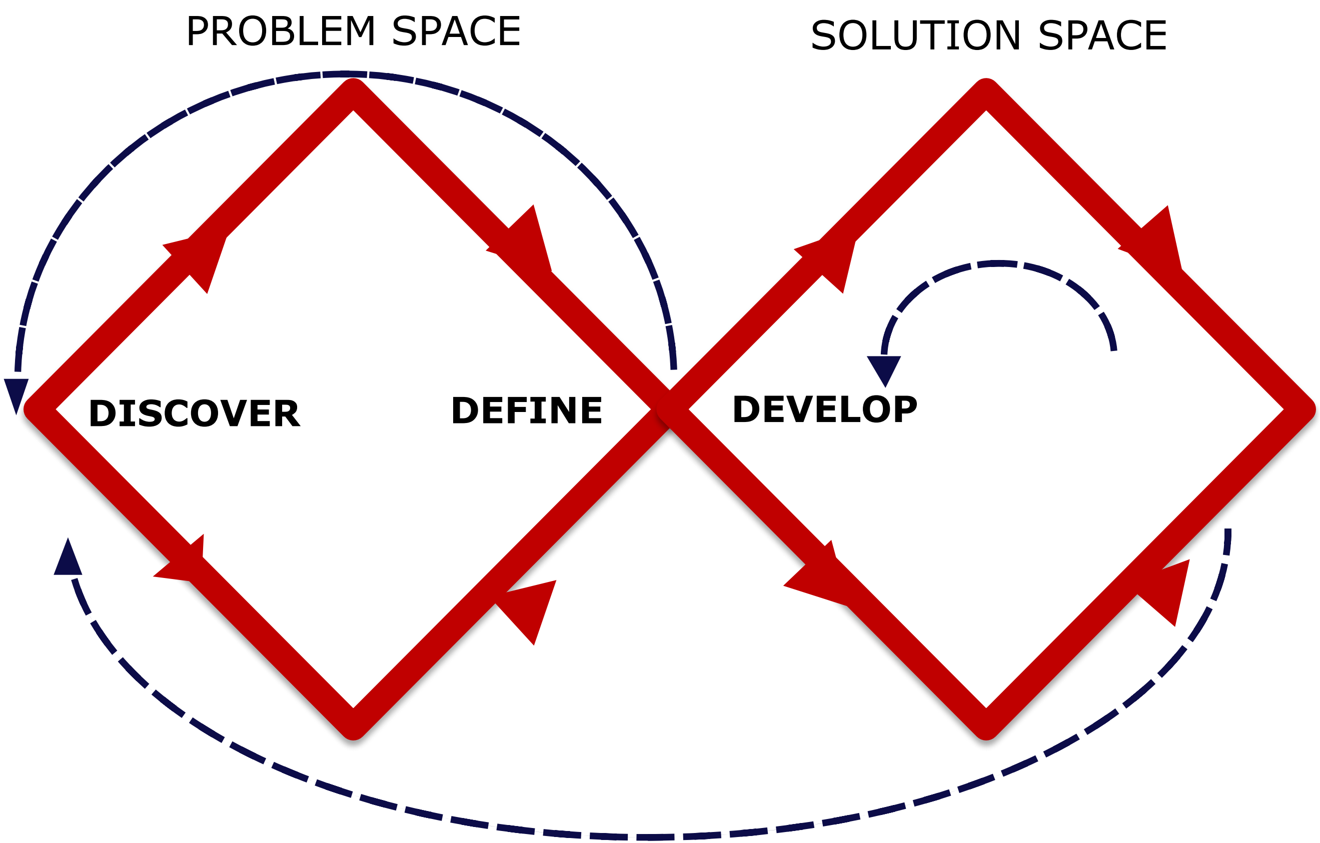
PS1

New Zealand's plastic waste management infrastructure is underdeveloped, leading to large amounts of plastic waste entering landfills and the environment, resulting in long term disruption to ecosystems

PS2

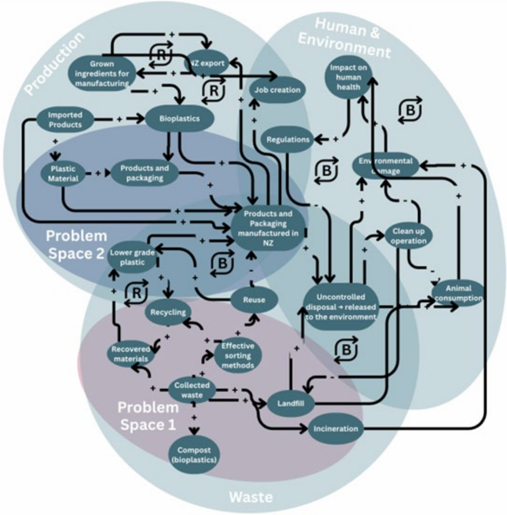
New Zealand's reliance on low grade plastics affect the environment due to their unsustainable production, use, disposal, and difficulty biodegrading

303 Recap: Divergent-Convergent Thinking



From Problem to Solution

Ideation → **Long list** → **Short list** → **Social CBA of the shortlist**



0 Do Nothing Do nothing, maintain the status quo.	**1 Plastic Alternatives** Replacing hard to recycle plastics with biodegradable alternatives such as bioplastics.
2 New Infrastructure Develop targeted recycling and composting infrastructure locally, to effectively process specific plastics and organic materials.	**3 Tax on Plastics** Tax on virgin and non-recyclable plastic packaging made from less than a certain percent recycled plastic.
4 Eliminating Microplastics Eliminating microplastics used as ingredients in everyday products e.g. cosmetics, to mitigate adverse human health effects. Multiple countries have already implemented this.	**5 Automated Plastic Sorters** Train a model to recognise plastic, paper, cans etc and let it sort these automatically. Would help speed up the recycling process. This technology will be used in common disposal locations, e.g. malls and supermarkets.
6 Public Bin Options Include a recycling option for all public bins.	**7 Educating Wider Community** Increase education and raise public awareness about consumer-stage separation of differing plastic types before putting rubbish into bins.
8 Improving Labelling Improve clarity and recycling packaging/labels to clearly disclose if it is recyclable and relevant environmental/health risks.	**9 Minimum Recycled Content** Enforce legally required percentages of post-consumer recycled (PCR) material in single-use plastic packaging.
10 Return System Introduce an incentivised recycling scheme, where consumer receive a small refund for returning type 1, 2, 5 plastic items for proper collection and recycling.	**11 Recycled Plastics Subsidy** Government subsidy for businesses to promote the use of recycled plastic as alternatives.

Short list

- Do nothing
- Option/Package 1
- Option/Package 2
- Option/Package 3

- Do nothing
- Do minimum
- Do medium
- Do maximum

Social CBA of the shortlist

- Net Present Value (NPV)
- Benefit Cost Ratio (BCR)
- Internal Rate of Return (IRR)

Covered next lecture

Ideation and generating options

PS1

New Zealand's plastic waste management infrastructure is underdeveloped, leading to large amounts of plastic waste entering landfills and the environment, resulting in long term disruption to ecosystems

PS2

New Zealand's reliance on low grade plastics affect the environment due to their unsustainable production, use, disposal, and difficulty biodegrading

2 New Infrastructure

Develop targeted recycling and composting infrastructure locally, to effectively process specific plastics and organic materials.

5 Automated Plastic Sorters

Train a model to recognise plastic, paper, cans etc and let it sort these automatically. Would help speed up the recycling process. This technology will be used in common disposal locations, e.g. malls and supermarkets.

10 Return System

Introduce an incentivised recycling scheme, where consumer receive a small refund for returning type 1, 2, 5 plastic items for proper collection and recycling.

7 Educating Wider Community

Increase education and raise public awareness about consumer-stage separation of differing plastic types before putting rubbish into bins.

8 Improving Labelling

Improve clarity and recycling packaging/ labels to clearly disclose if it is recyclable and relevant environmental/health risks.

1 Plastic Alternatives

Replacing hard to recycle plastics with biodegradable alternatives such as bioplastics.

3 Tax on Plastics

Tax on virgin and non-recyclable plastic packaging made from less than a certain percent recycled plastic.

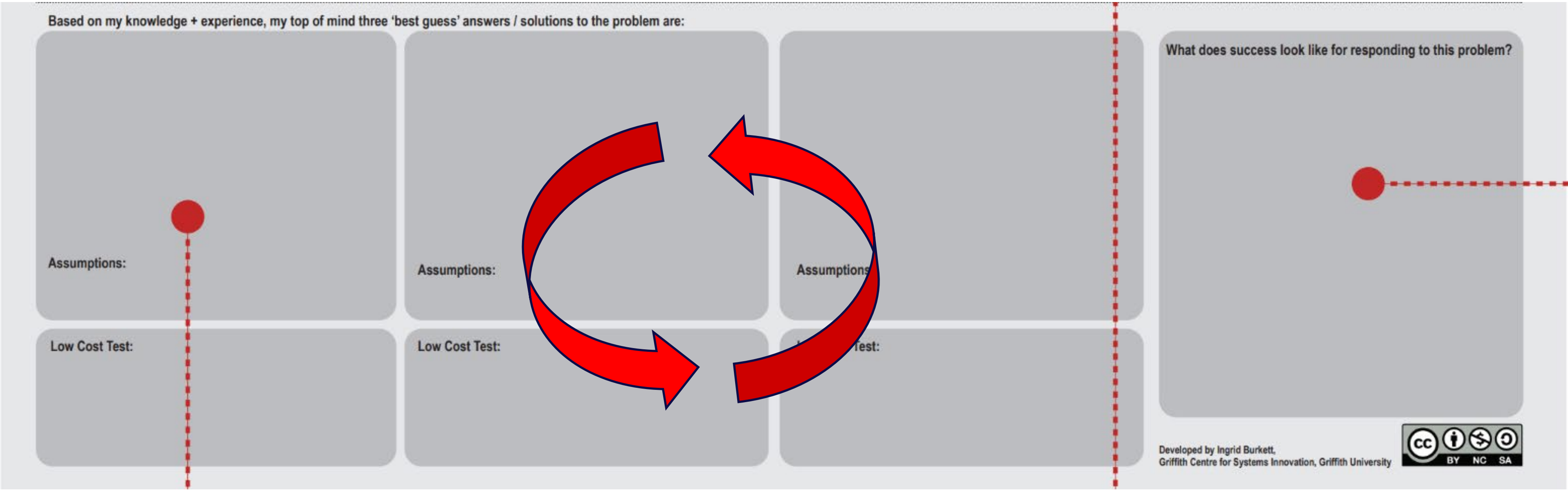
4 Eliminating Microplastics

Eliminating microplastics used as ingredients in everyday products e.g. cosmetics, to mitigate adverse human health effects. Multiple countries have already implemented this.

0 Do Nothing

Do nothing, maintain the status quo.

Ideation and generating options



What are your assumptions underpinning 'guesses'?

Low cost test: How valid are your assumptions, are you addressing the PS?

What does success look like? Imagine what it would be like if the problem wasn't there

Options assessment of the long list – Critical Success Factors

Key Stakeholder
Ministry for the Environment
Ministry of Foreign Affairs and Trade
Plastic Regulators
Recycling Facilities
Plastic Producers
Local Iwi / Māori Representatives
Other Key Stakeholders

CSF 1
Regulatable through accurate & relevant measurements

CSF 2
Long-term waste management operations are affordable & sustainable

CSF 3
Does not conflict with iwi cultural values in waste management practices

CSF 4
Adjusts economic incentives to promote comprehensive behavioural change

CSF 5
Minimise additional financial strain on communities & businesses

CSF 6
Reduces the adverse effects of plastic on the environment

Options assessment of the long list – DFV analysis

Desirability	Feasibility	Viability
Are we solving the right problem for the right people?	**Can this solution realistically be implemented?**	**Can this solution deliver long-term value and be sustained?**
Stakeholder needs and priorities	Technical capability	Economic sustainability
Equity, wellbeing, and environmental outcomes	Infrastructure and existing systems	Funding and investment
Trade-offs between competing group interests	Policy, regulation, and governance	Long-term operating and maintenance costs
Social acceptance and political support	Organisational capability and collaboration	Scalability and resilience
Unintended consequences across the system	Dependencies between interacting systems	Benefits outweigh costs over the system lifecycle

Options assessment of the long list – DFV analysis

Option	Desirability	Feasibility	Viability
Ban single-use plastics	Reduces litter and environmental harm. May inconvenience consumers and businesses.	Requires legislation, enforcement, and suitable alternatives.	Long-term environmental benefits, but potential increased costs and political resistance.
Invest in recycling infrastructure	Reduces waste sent to landfill and recovers valuable materials.	Requires significant investment, technology, and public participation.	Sustainable if recycling markets remain profitable and contamination is kept low.
Introduce a plastic tax	Encourages consumers and businesses to reduce plastic use. May disproportionately affect low-income households.	Relatively easy to implement through existing taxation systems.	Generates government revenue and incentivises behaviour change, but effectiveness depends on tax level and public acceptance.

Options assessment

Table 1: *DFV and CSF analysis of longlisted options*

Options Long-List	D	F	V	CSF 1	CSF 2	CSF 3	CSF 4	CSF 5	CSF 6
0									
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									

Options assessment of the long list – Included vs excluded options

Recycling and composting infrastructure (Option 2), a tax on new and non-recyclable plastics (Option 3), improving recycling labelling (Option 8), recycling return systems (Option 10), and recycled plastic subsidies (Option 11) were carried forward as they met both the DFV requirements and identified CSFs. Public bin options (Option 6) met all but CSF 1 and 4, and educating the wider community (Option 7) met all but CSF 4, as they are not sustainable on their own and do not meet economic targets. However, these limitations can be mitigated when paired with other options that offer stronger regulatory and economic measures. As a result, both Option 6 and 7 serve as valuable complementary components within broader solution packages.

Plastic alternatives (Option 1) and eliminating microplastics (Option 4) were removed as they were not considered to be viable, costing $100 million to $1 billion and $10 billion to $20 billion over 10 years respectively, likely exceeding the prescribed budget (Ministry for the Environment, 2022; Kent, 2019; Doyle, 2022). Both options will cause increased costs to the supply chain and have the potential to cause costs to consumers, failing CSF 5. Additionally, Option 4 only considers microplastics which accounts for a small percentage of plastic leakage. Therefore, benefits to the environment are limited, and it does not perfectly align with CSF 6.

Shortlist packages

Once you have analysed your longlist you will be able to narrow options to a shortlist

The preferred solution is often a package of interventions that work together to address the system as a whole.

Table 11: CSF Analysis of Do Medium Option

Options Long-List	Option	CSF 1	CSF 2	CSF 3	CS4	CSF 5	CSF 6
3	Tax on Plastics						
10	Return System						
11	Recycled Plastics Subsidy						

Satisfies

Partially Satisfies

Does not Satisfy

Shortlist packages

The *Do Minimum* case offers modest reductions in methane emissions and cost savings from landfill diversion. However, while it achieves a positive NPV of $34 million and a BCR of 1.51, its scope is limited to benefits that largely rely on consumer behaviour through education campaigns, advertising, and small-scale bin rollouts. Therefore, systemic issues of plastic overconsumption, low recovery rates, and reliance on virgin plastics remain largely unaddressed.

The *Do Maximum* case promises the greatest environmental and economic gains through extensive investment in composting and recycling facilities, subsidies for bioplastics, and high taxation on virgin plastics. However, this package involves extremely high upfront and ongoing costs which would place a heavy burden on government finances, as well as high taxes that place pressure on businesses. This poses significant implementation risks within the 10-year horizon, making it unsuitable in the current economic climate.

The *Do Medium* case strikes the most effective balance. By introducing the VPPT alongside the Incentivised Return Scheme, supported by 2400 Reverse Vending Machines, this option creates both economic and social incentives for behavioural change. An NPV of $237 million and a BCR of 1.74 for the *Do Medium* case is financially sustainable and generates significant quantifiable benefits across the LSF capitals. Environmental benefits include a $111 million reduction in costs relating to CO_2 emissions and $48 million in health-related benefits from reduced methane emissions. Socially, the scheme strengthens community engagement and builds human capital through education and recycling participation.

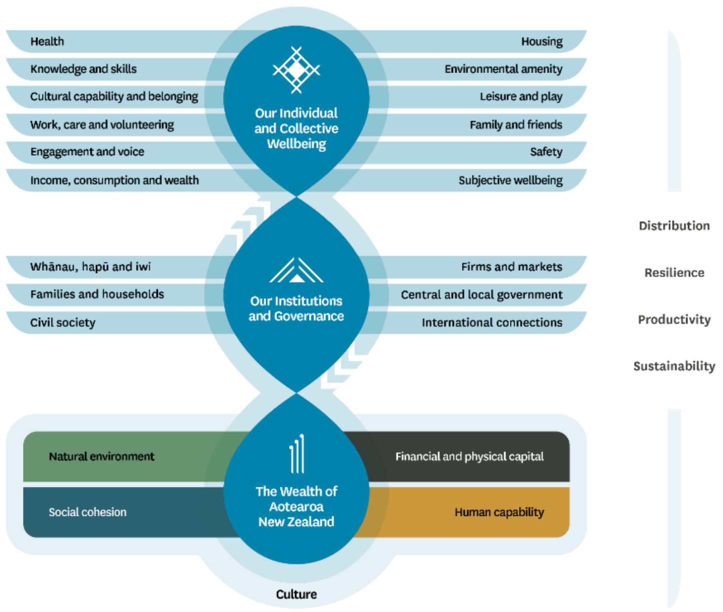
Economic and social impact assessment

	Do Nothing	Do Minimum	Do Medium	Do Maximum
Net Present Value (NPV)	-$19.9 billion	$34 million	$237 million	$222 million
Benefit Cost Ratio (BCR)	N/A	1.51	1.74	1.38
Internal Rate of Return (IRR)	N/A	36%	85%	30%

Carry out Social CBA, and social impact assessment (qualitative) on shortlist packages

More on this in lecture 5 and 6 this week

The Treasury's Living Standards Framework



Preferred way forward

Overall, the *Do Medium* case represents the most balanced and achievable pathway. It addresses key environmental and health impacts more effectively than the *Do Minimum* option, while avoiding the prohibitive costs and risks of the *Do Maximum* package. This package also aligns with broader government strategies on waste minimisation, circular economy development, and climate change mitigation, as well as aligning with the CSF’s seen in Table 11.

Table 11: CSF Analysis of Do Medium Option

Options Long-List	Option	CSF 1	CSF 2	CSF 3	CS4	CSF 5	CSF 6
3	Tax on Plastics						
10	Return System						
11	Recycled Plastics Subsidy						

Satisfies

Partially Satisfies

Does not Satisfy

Business case limitations

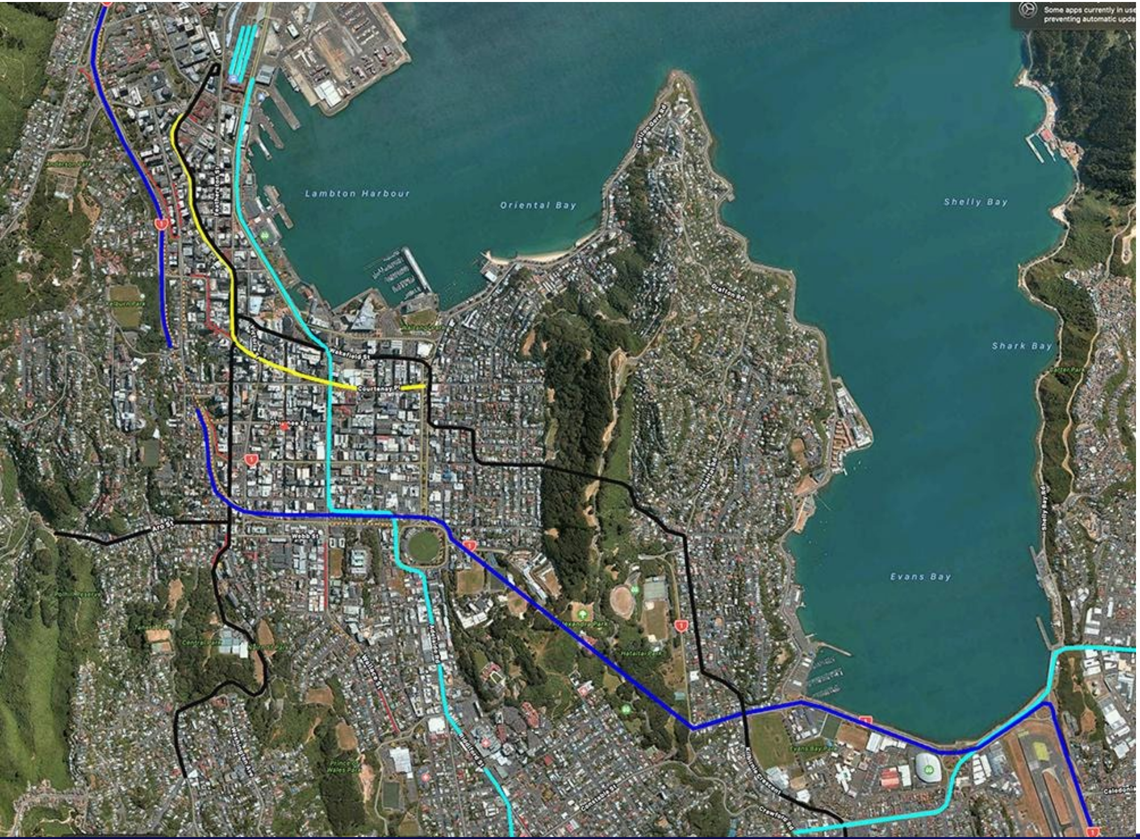
- Relies heavily on assumptions and forecasts (things change)
- Cannot capture every social or environmental impact
- Simplifies complex systems and interactions
- May overlook unintended consequences and trade offs
- Decision making also involves politics, values, and judgement

New Waikato Medical School

New Waikato Business School documents - Ministry of Health

Flaws in Waikato med school business case
– Otago Daily Times

Lets Get Wellington Moving Program



Scrapped in [2023]

- Long delivery timeframes
- Changing political priorities
- Governance complexity
- Stakeholder disagreement
- Evolving public expectations
- Scrutiny around delivery of benefits and project costs

Questions?

Thank you, and see you next lecture ☺